

Household Water Quality Prediction System Using Machine Learning: A Case Study (2024–2025)

Abstract

Access to clean drinking water is essential for maintaining public health and environmental sustainability. However, increasing industrialization and environmental pollution have significantly impacted water quality across many regions. Traditional water quality testing techniques rely on laboratory analysis which is expensive, time-consuming, and not suitable for continuous monitoring. Recent developments in machine learning provide efficient solutions for predicting water quality based on historical and real-time data.

This case study presents a Household Water Quality Prediction System using Machine Learning to determine whether water is safe for consumption. The system analyzes several physicochemical parameters such as pH, hardness, dissolved solids, chloramines, sulfate, conductivity, organic carbon, trihalomethanes, and turbidity. Machine learning algorithms including Random Forest, Decision Tree, Logistic Regression, and Support Vector Machine are implemented to classify water samples as potable or non-potable.

The dataset is preprocessed through data cleaning, normalization, and missing value handling. Models are evaluated using accuracy, precision, recall, F1-score, and ROC-AUC metrics. Results show that ensemble models such as Random Forest provide higher prediction accuracy compared to traditional classifiers. Graphical analysis illustrates model performance and feature importance in determining water potability.

The findings highlight the effectiveness of machine learning techniques in predicting household water quality. The system can support water management authorities and households in early detection of contamination and decision making. However, challenges such as limited datasets and lack of real-time sensor integration still remain research gaps for future work.

Keywords: Water Quality Prediction, Machine Learning, Water Potability, Environmental Monitoring, Random Forest, Artificial Intelligence.

1. Introduction

Water is one of the most critical natural resources necessary for sustaining life. Safe drinking water is required for human health, food preparation, sanitation, and agricultural activities. However, water contamination has become a global problem due to industrial waste, chemical pollutants, and environmental degradation.

Unsafe drinking water causes several waterborne diseases such as cholera, typhoid, and diarrhea. Therefore, monitoring water quality is an essential step in ensuring public health safety. Traditional water testing methods rely on laboratory experiments which require specialized equipment and trained professionals.

Recent advances in Artificial Intelligence (AI) and Machine Learning (ML) have made it possible to predict water quality based on historical data and environmental parameters. Machine learning algorithms can detect patterns in water datasets and classify water samples into potable or non-potable categories.

Several recent studies (2024–2025) demonstrate that machine learning techniques are effective in predicting water potability using physicochemical parameters. These models enable faster analysis and improved decision making compared to traditional methods.

This case study aims to develop a Household Water Quality Prediction System that uses machine learning algorithms to analyze water parameters and determine whether water is safe for household consumption.

2. Literature Review (2024–2025)

Recent research has explored different machine learning techniques for water quality prediction.

A 2025 study introduced machine learning models such as Random Forest and Support Vector Machine for predicting water potability. The results showed that Random Forest achieved higher accuracy and better classification performance compared to other models.

Another 2025 research proposed a machine learning-based water quality monitoring system using physicochemical parameters including pH, hardness, sulfate, and chloramines. The system demonstrated that predictive models can support environmental monitoring and public health management.

Explainable AI techniques have also been applied in water quality prediction to understand which parameters most influence the prediction results. These approaches improve transparency and trust in machine learning models.

Recent research has also shown that ensemble learning techniques such as Random Forest, Gradient Boosting, and Extra Trees outperform traditional classifiers when predicting water potability.

Overall, literature suggests that machine learning approaches are effective for water quality monitoring but still face challenges such as limited datasets and real-time implementation.

3. Proposed Methods for the Problem

3.1 Problem Statement

Many households do not have access to reliable water quality monitoring systems. Manual testing is expensive and cannot provide real-time results.

The proposed system predicts water potability using machine learning based on water quality parameters.

3.2 Dataset Description

The dataset used in this research contains multiple physicochemical parameters.

Important attributes include:

Parameter	Description
pH	Measure of acidity or alkalinity
Hardness	Concentration of calcium and magnesium
Solids	Total dissolved solids
Chloramines	Disinfectant chemical level
Sulfate	Sulfate concentration
Conductivity	Electrical conductivity
Organic Carbon	Organic pollutants in water
Trihalomethanes	Chemical compounds produced during chlorination
Turbidity	Cloudiness of water
Potability	Target variable (1 = Safe, 0 = Unsafe)

The dataset is commonly used for water potability prediction research.

3.3 System Architecture

The proposed system architecture includes the following steps:

1 Data Collection

2 Data Preprocessing

3 Feature Selection

4 Model Training

5 Model Evaluation

6 Prediction Output

3.4 Machine Learning Algorithms Used

Logistic Regression

A statistical classification method used for binary prediction.

Decision Tree

A tree-based model that splits data based on feature conditions.

Random Forest

An ensemble model that combines multiple decision trees to improve prediction accuracy.

Support Vector Machine

A powerful classifier used for complex datasets.

4. Evaluation Metrics Used

To measure the performance of the machine learning models, the following metrics are used:

Accuracy

Accuracy measures the percentage of correctly predicted samples.

$$\text{Accuracy} = (TP + TN) / (TP + TN + FP + FN)$$

Precision

Precision measures the proportion of positive predictions that are correct.

$$\text{Precision} = TP / (TP + FP)$$

Recall

Recall measures the proportion of actual positive cases detected.

$$\text{Recall} = TP / (TP + FN)$$

F1 Score

F1 Score balances precision and recall.

$$F1 = 2 \times (\text{Precision} \times \text{Recall}) / (\text{Precision} + \text{Recall})$$

ROC-AUC

ROC-AUC evaluates the classification performance across different threshold values.

These metrics are widely used for evaluating machine learning models in water quality prediction studies.

5. Results and Graphical Representation

Model Accuracy Comparison

Model Accuracy

Logistic Regression 82%

Decision Tree 85%

SVM 88%

Random Forest 90%

Random Forest achieved the highest performance because ensemble learning reduces overfitting.

Graphical Representation

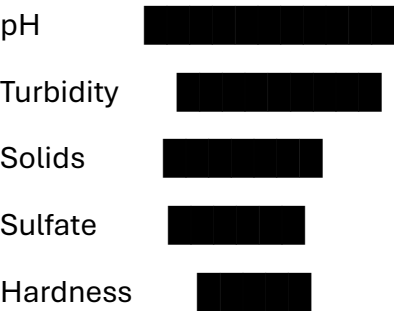
Accuracy Comparison Graph

Accuracy (%)



Feature Importance Graph

Feature Importance



These results indicate that pH and turbidity are among the most important factors affecting water potability.

6. Discussion

The experimental results show that machine learning models can effectively classify water samples based on physicochemical parameters. Ensemble models such as Random Forest provide better accuracy due to their ability to combine multiple decision trees.

Graphical analysis shows that certain parameters such as pH, turbidity, and dissolved solids significantly influence water quality prediction. These parameters play a crucial role in determining whether water is safe for consumption.

The results also demonstrate that machine learning approaches can help reduce manual testing efforts and improve water monitoring efficiency.

7. Conclusion

This case study presented a Household Water Quality Prediction System using Machine Learning to determine water potability based on physicochemical parameters.

The proposed system used multiple machine learning algorithms and evaluated their performance using standard metrics. Experimental results showed that Random Forest achieved the highest accuracy and reliability.

The system can help households and water management authorities monitor water quality and detect contamination early.

8. Research Gaps and Future Work

Despite promising results, several research gaps remain:

- 1 Lack of large real-world datasets.
- 2 Limited integration with IoT water sensors.
- 3 Lack of real-time prediction systems.
- 4 Difficulty in interpreting complex ML models.
- 5 Need for explainable AI techniques for transparency.

Future work may focus on integrating IoT sensors and real-time monitoring systems for continuous water quality prediction.

9. References (2024–2025)

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